Results pack Q5: Fluid balance and ICU length of stay in survivors

FLUID-ICU is a SIMULATED dataset made for teaching. No real patients. Neudata | Foundations of Clinical Research, Level 1

SIMULATED DATA: NOT REAL PATIENTS. FLUID-ICU was generated by computer for this course. Write your manuscript as a training exercise and state clearly that the data are simulated.

1. The question

Among adults with sepsis who survived to day 28, is a positive cumulative fluid balance by day 3 associated with a longer ICU stay?

2. Design and data

- Design: multicentre prospective cohort study (observational); analysis restricted to 28-day survivors.
- Setting: 4 ICUs (ICU A-D), consecutive adults with sepsis, enrolment January 2023 to December 2024.
- Exposure: positive vs negative cumulative fluid balance at day 3; secondary: per 10 ml/kg.
- Outcome: ICU length of stay in days (icu_los_days), right-skewed.
- Analysis sample: 439 survivors with a day-3 balance (239 positive, 200 negative); adjusted model 366 complete cases.
- Pre-specified confounders (SAP): age, SOFA score, lactate, vasopressor, mechanical ventilation, chronic kidney disease.
- Software: R 4.6 (packages gtsummary, survival, mice). Scripts: Course/Data/R/.

3. Table 1: who was studied

Table 1. Baseline characteristics of 28-day survivors by fluid-balance group

Characteristic	Overall N = 439[1]	Negative balance N = 200[1]	Positive balance N = 239[1]
Age, years	61.3 (14.7)	63.8 (14.6)	59.1 (14.6)
Missing	1	1	0
Sex			
Female	178 (41%)	85 (43%)	93 (39%)
Male	261 (59%)	115 (58%)	146 (61%)
Weight, kg	64.5 (11.8)	64.7 (12.5)	64.2 (11.3)
Diabetes	108 (25%)	54 (27%)	54 (23%)
Chronic kidney disease	33 (7.5%)	16 (8.0%)	17 (7.1%)
Heart failure	60 (14%)	43 (22%)	17 (7.1%)
Hypertension	196 (45%)	102 (51%)	94 (39%)
Source of infection			
Respiratory	191 (44%)	84 (42%)	107 (45%)
Urinary	99 (23%)	45 (23%)	54 (23%)
Abdominal	71 (16%)	37 (19%)	34 (14%)

Characteristic	Overall N = 439[1]	Negative balance N = 200[1]	Positive balance N = 239[1]
Skin/soft tissue	36 (8.2%)	15 (7.5%)	21 (8.8%)
Other	42 (9.6%)	19 (9.5%)	23 (9.6%)
SOFA score	7 (5, 9)	6 (4, 7)	8 (5, 10)
APACHE II score	18 (13, 22)	16 (12, 20)	19 (15, 23)
Missing	5	3	2
Lactate, mmol/L	2.3 (1.6, 3.4)	2.1 (1.4, 2.9)	2.6 (1.8, 3.6)
Missing	73	45	28
Mean arterial pressure, mmHg	75 (68, 82)	78 (71, 85)	73 (67, 79)
Missing	3	1	2
Vasopressor on admission	152 (35%)	39 (20%)	113 (47%)
Mechanical ventilation	189 (43%)	67 (34%)	122 (51%)

[1]Mean (SD); n (%); Median (Q1, Q3)

Mean (SD); n (%); median (Q1, Q3). Missing = number of patients without that value.

4. Main results

- ICU stay is right-skewed: in survivors mean 7.5 vs median 6.2 days.
- Median (IQR) ICU stay: positive balance 7.5 (5.2 to 11.2) vs negative 4.8 (3.3 to 7.1) days; Mann-Whitney p <0.001.
- Hodges-Lehmann difference: 2.5 (95% CI 1.8 to 3.2) days.
- Log-linear regression: crude ratio of geometric means 1.50 (95% CI 1.34 to 1.68); adjusted **1.24 (95% CI 1.10 to 1.39)**, p <0.001 (i.e. about 24% longer stay).
- Per 10 ml/kg more balance: ratio 1.06 (95% CI 1.03 to 1.08).

Table 2. ICU length of stay in survivors by fluid-balance group

Summary	Negative balance	Positive balance
n	200	239
Median (IQR), days	4.8 (3.3 to 7.1)	7.5 (5.2 to 11.2)
Mean (SD), days	6.0 (4.6)	8.7 (5.0)
Range, days	1.2 to 41.9	1.2 to 27.9
Geometric mean, days	4.9	7.4

Table 3. Association between fluid balance and ICU length of stay (survivors)

Analysis	n	Estimate (95% CI)	p
Mann-Whitney (Hodges-Lehmann difference, days)	439	2.5 (1.8 to 3.2)	<0.001
Log-linear, crude (ratio of geometric means)	439	1.50 (1.34 to 1.68)	<0.001
Log-linear, adjusted (ratio of geometric means)	366	1.24 (1.10 to 1.39)	<0.001

Analysis	n	Estimate (95% CI)	p
Log-linear, adjusted, per 10 ml/kg (ratio)	366	1.06 (1.03 to 1.08)	<0.001

Ratio of geometric means: 1.24 = 24% longer typical stay. Adjusted for age, SOFA, lactate, vasopressor, ventilation, CKD.

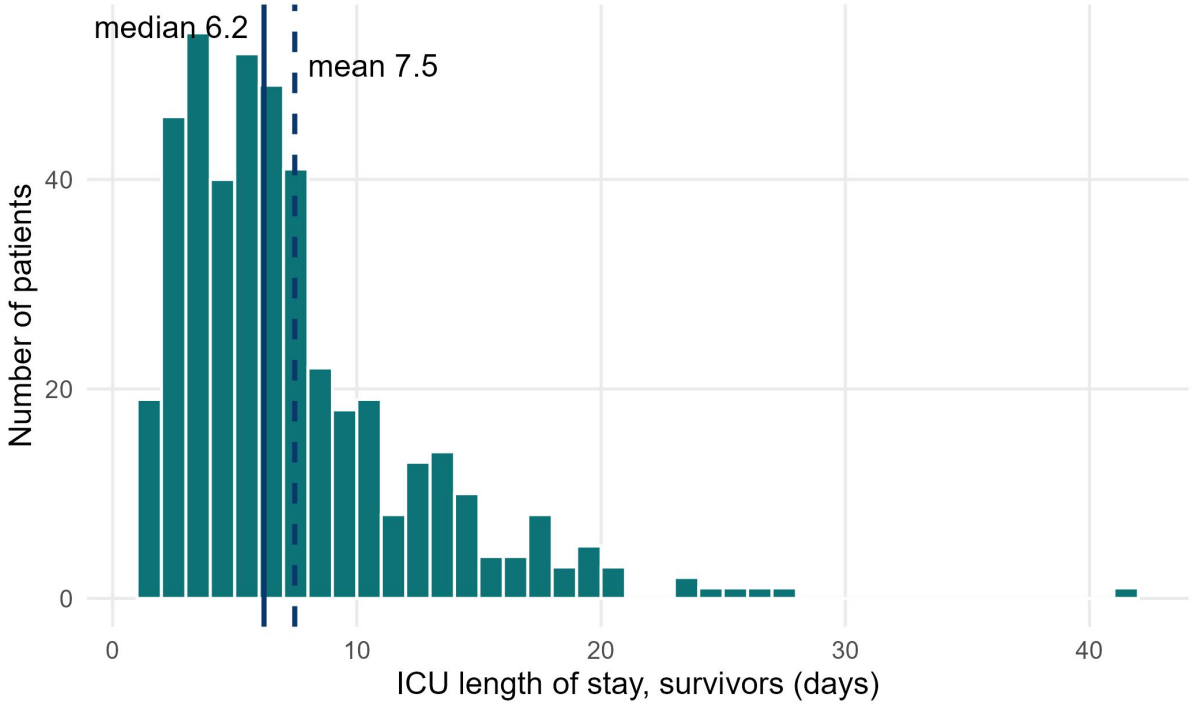


Figure 1. ICU length of stay in survivors: a long right tail pulls the mean above the median

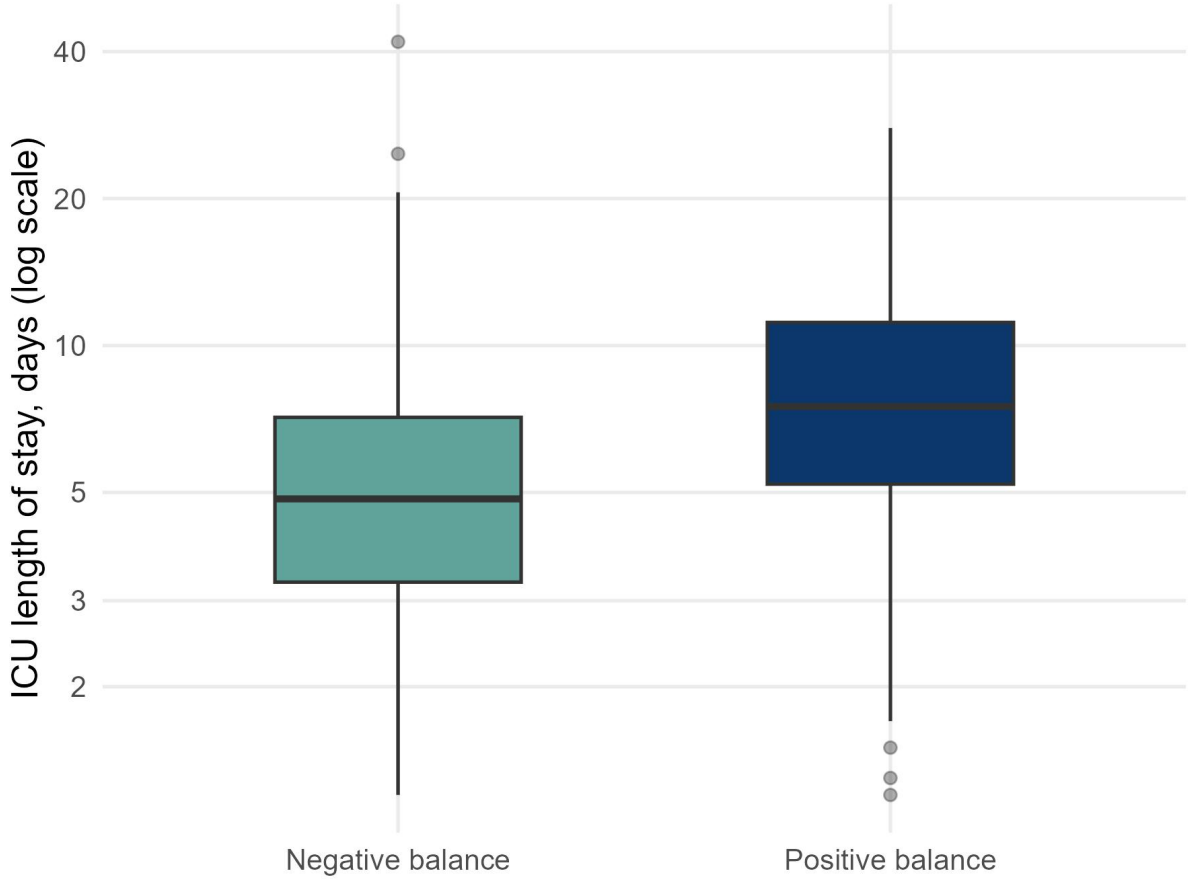


Figure 2. ICU length of stay by fluid-balance group (log scale)

5. What this means (plain language)

- Among survivors, those with a positive balance stayed longer in the ICU, by about 2-3 days in the median.
- After adjustment, the typical (geometric mean) stay was about 24% longer.
- For a skewed outcome, report the median (IQR), use a rank test or analyse the log of the stay, and present a ratio.

6. What this does NOT mean

- It does NOT show that fluid prolongs ICU stay; patients who stay longer have more time and reasons to receive fluid (reverse causation).
- It does NOT describe all patients: deaths were excluded, and death shortens the stay.

7. Limitations to discuss (hints)

- Selection: restricting to survivors conditions on an outcome that is itself related to fluid (survivor bias). An alternative is 'ICU-free days', which counts death as 0.
- ICU discharge depends on bed pressure and local practice, not only on the patient.
- Residual confounding by severity; complete-case adjusted model.

Which STROBE items this pack helps you write

- Item 6: participants (why survivors only)
- Item 12: methods for a skewed outcome (log transformation / non-parametric)
- Item 14: descriptive data (Table 1)
- Item 15-16: medians (IQR) and ratios with 95% CI
- Item 19: limitations (survivor selection)